

Neuromorphic ASR

Transducer Models on Memristor Hardware

Azim Javed

`azim.javed@ml.rwth-aachen.de`

August 22, 2026

Advisor: Dr. Albert Zeyer

Supervisor: Prof. Ralf Schlüter

Human Language Technology and Pattern Recognition
Computer Science Department, RWTH Aachen University

Outline

Related Work

Motivation

Prerequisites

Experiments

Future Work

Outline

Related Work

Motivation

Prerequisites

Experiments

Future Work

Related Work

- ▶ Memristive crossbars enable analog in-memory matrix–vector multiplication, promising energy-efficient inference for large neural networks [Strukov & Snider⁺ 08, Aguirre & Sebastian⁺ 24].
- ▶ Most prior memristor ML studies are limited to small networks and tasks (MNIST-level, keyword spotting); common simulators (MemTorch, IBM AIHWKIT) rely on idealized hardware assumptions [Rossenbach & Hilmes⁺ 25].
- ▶ Rossenbach et al. [Rossenbach & Hilmes⁺ 25] (Interspeech 2025):
 - ▷ First full-scale ASR system (Conformer CTC, TED-LIUMv2) executed on realistically simulated memristor hardware.
 - ▷ Simulation via Synaptogen [Hennen & Brackmann⁺ 24], a device model fitted to measured programming statistics.
 - ▷ With QAT down to 3-bit weights: only $\sim 25\%$ relative WER degradation on device.
- ▶ Open questions left by [Rossenbach & Hilmes⁺ 25]: transducer topologies and recurrent components on device — the subject of this thesis.

Outline

Related Work

Motivation

Prerequisites

Experiments

Future Work

Motivation

- ▶ Starting point: CTC-based Conformer ASR has been shown to run on (simulated) memristor hardware [Rossenbach & Hilmes⁺ 25].
- ▶ Curiosity: how does a **transducer** model, architecturally more involved than CTC, behave on memristor hardware?
 - ▷ Autoregressive prediction network with an internal language model (ILM).
 - ▷ Joint network evaluated repeatedly per emitted label.
- ▶ Study the impact of offloading various components of the transducer to the device (encoder only vs. encoder + prediction network).
- ▶ Study the actual impact of device noise, beyond pure quantization.
- ▶ Performance impact on lexicon-free vs. tree-based searches.
- ▶ LM point of view: how does device deployment change the interplay between prior/ILM correction and the external LM?

Outline

Related Work

Motivation

Prerequisites

Experiments

Future Work

Memristors and Crossbar Arrays

- ▶ Memristor: passive two-terminal device with non-volatile, programmable resistance [Strukov & Snider⁺ 08]. A weight is stored as a conductance $G = 1/R$.
- ▶ Crossbar array: wordlines (rows) $\times$ bitlines (columns), one memristor per junction.
 - ▷ Inputs applied as voltages V_j on the wordlines.
 - ▷ By Ohm's law and Kirchhoff's current law, each bitline aggregates a column:
$$I_i = \sum_j G_{ij} V_j.$$
 - ▷ Matrix–vector multiplication happens in one physical step, in parallel over all outputs.
- ▶ Signed weights via pairwise encoding [Aguirre & Sebastian⁺ 24]:
 $w_{ij} = G_{ij}^+ - G_{ij}^-$ on two bitlines (an $N \times M$ matrix needs an $N \times 2M$ crossbar).
- ▶ Only binary programming is precise $\Rightarrow$ multi-bit weights via bit-stacking: a k -bit weight is split over $k-1$ stacked crossbars (4-bit: levels $-7, \dots, 7$).
- ▶ Inputs/outputs pass 8-bit DAC/ADC converters (input range ± 0.6 V).

Non-Idealities and Synaptogen

- ▶ Ideal crossbars are assumed by many simulators; real devices exhibit:
 - ▷ Device-to-device and cycle-to-cycle programming variability,
 - ▷ Thermal (Johnson–Nyquist) noise,
 - ▷ Finite DAC/ADC resolution,
 - ▷ Imprecise intermediate (non-binary) conductance levels.
- ▶ Synaptogen [Hennen & Brackmann⁺ 24]: generative device model based on a vector autoregression, fitted to 3M+ measured programming cycles of 130 nm memristors (STMicroelectronics).
- ▶ In our experiments (PyTorch, SynaptogenML): linear layers are replaced by crossbar simulation, the rest of the network stays conventional.

Table: Single crossbar multiplication for target value $1.0 \times X$, over differently programmed devices [Rossenbach & Hilmes⁺ 25].

Target	avg	std	min	max
1.0×1	1.016	0.063	0.707	1.270
1.0×0.5	0.476	0.094	0.109	1.060
1.0×0	$8.5 \cdot 10^{-4}$	$4.8 \cdot 10^{-2}$	-0.275	0.246

Quantization-Aware Training

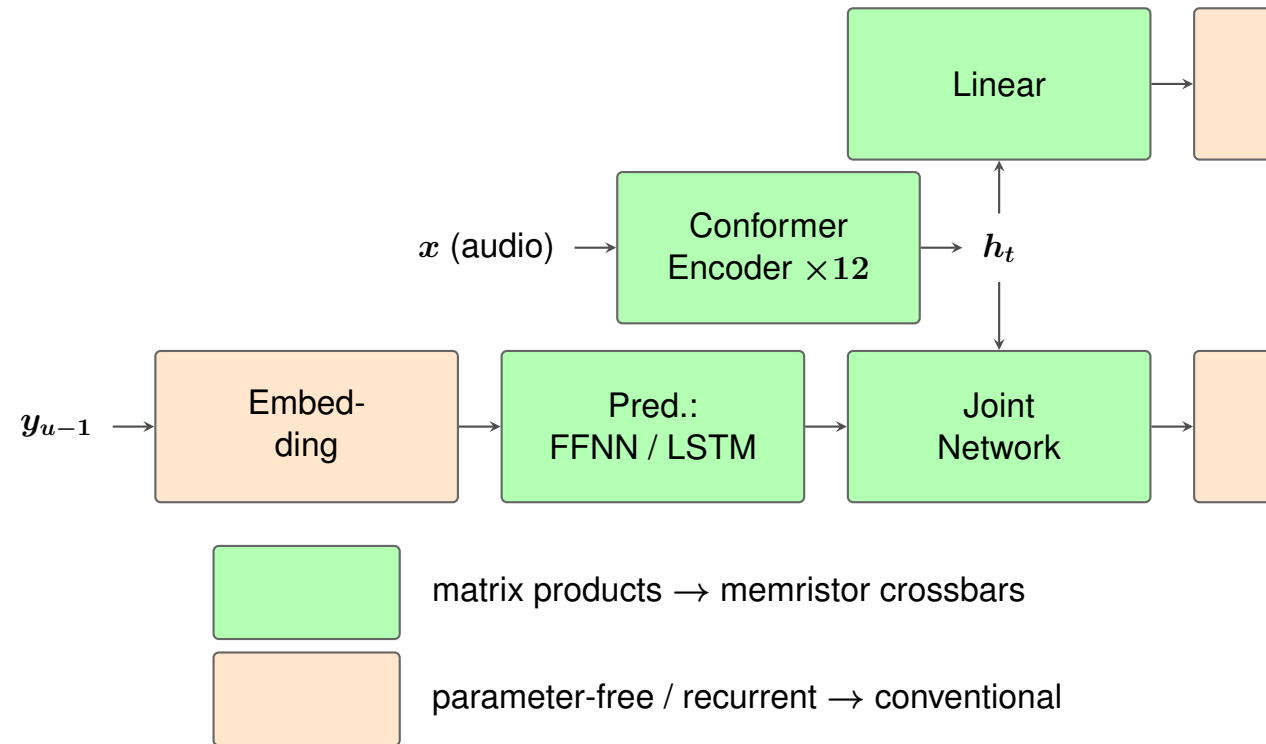
- ▶ Memristor execution effectively means: low-precision (stacked binary) weights + 8-bit DAC/ADC on activations.
- ▶ Post-training quantization (PTQ) collapses at low precision; quantization-aware training (QAT) [Gholami & Kim⁺ 21] inserts fake quantization into the training from epoch 1.
 - ▷ Symmetric per-tensor quantization, zero-point fixed to 0 (required by the pairwise encoding).

Table: WER (%) on TED-LIUMv2 dev, Conformer CTC, 4-gram LM [Rossenbach & Hilmes⁺ 25].
Memristor: mean $\pm$ std over 10 differently programmed devices.

Weight bits	FP32	8	6	5	4	3	2
PTQ	7.2	7.2	7.9	8.9	11.4	30.7	–
QAT		7.3	7.7	7.4	7.8	8.3	22.1
Memristor (simulated)		8.3	8.3	8.3	8.9	9.2	–

- ▶ Up to 3-bit weights: $\leq 25\%$ relative WER degradation on device (9.2 vs. 7.2).

Model Topologies: What Can Run on Device?



- ▶ CTC [Graves & Fernández⁺ 06]: encoder only, label-synchronous decoding, conditionally independent outputs.
- ▶ Transducer [Graves 12]: encoder + prediction network + joint network, autoregressive over labels.
- ▶ All linear (matmul) layers map to **crossbars (green)**.
- ▶ Parameter-free parts stay

Outline

Related Work

Motivation

Prerequisites

Experiments

Future Work

Experimental Setup

- ▶ Data: LibriSpeech [Panayotov & Chen⁺ 15] (960 h), BPE subword targets (128 merges → 184 labels).
- ▶ Features: 80 log-mel filters, 10 ms frame shift; SpecAugment during training.
- ▶ Encoder (shared by all models): 12-layer Conformer [Gulati & Qin⁺ 20], 512-dim.
- ▶ Model types:
 1. **CTC** [Graves & Fernández⁺ 06]: encoder only.
 2. **FFNN-transducer**: prediction network is a 2-layer feed-forward net (640-dim, tanh) over the previous-label embedding.
 3. **Full-context transducer** [Graves 12]: LSTM prediction network (standard RNN-T).
- ▶ Quantization: QAT, per-tensor symmetric, e.g. 4-bit weights / 8-bit activations; zero-point 0.
- ▶ Stages: (1) FP32 baseline → (2) QAT (fake quantization) → (3) memristor execution (Synaptogen crossbar simulation, multiple re-programmings per run).

Experiment Matrix

- ▶ Full matrix: 3 (model types) $\times$ 5 (search types) $\times$ 3 (stages).
- ▶ Search types:
 1. Lexicon-free, no LM
 2. Lexicon-free + BPE LSTM LM
 3. Prefix-tree beam search, no LM
 4. Prefix-tree beam search + 4-gram LM
 5. Prefix-tree beam search + BPE LSTM LM
(+ large-beam tree variants, beam 1024/2048)
- ▶ Prior/ILM correction and external LM scales are swept at decoding time.
- ▶ Reporting:
 - ▷ Primary tables: one per model type (search types $\times$ stages).
 - ▷ Appendix: tables across models at the same stage, for easy comparison.
- ▶ Expectation: device deployment changes the interplay between prior/ILM correction and the LM — quantization and noise reshape the acoustic scores, so the optimal LM weighting should shift.

Outline

Related Work

Motivation

Prerequisites

Experiments

Future Work

Future Work

- ▶ Error analysis: which error types does device noise introduce or amplify?
- ▶ Countermeasures: denoising (e.g. MLP), search parameters, LM parameters, softmax temperature.
- ▶ Fine-tuning of the (quantized) model; effect of noise introduction during training.
- ▶ Generalization: comparison across different datasets / data conditions.

Summary

- ▶ Extend memristor-based ASR [Rossenbach & Hilmes⁺ 25] from CTC to transducer models.
- ▶ Full matrix on LibriSpeech: 3 model types $\times$ 5 search types $\times$ 3 stages (baseline / QAT / memristor).
- ▶ Key questions: component offloading (encoder vs. encoder + prediction), real noise impact, lexfree vs. tree search behavior, prior/ILM–LM interplay.

Thank you for your attention!

Reference

- 📄 F. Aguirre, A. Sebastian, M. Le Gallo et al.
Hardware Implementation of Memristor-Based Artificial Neural Networks.
Nature Communications, Vol. 15, pp. 1974, March 2024.
- 📄 A. Gholami, S. Kim, Z. Dong, Z. Yao, M. W. Mahoney, K. Keutzer.
A Survey of Quantization Methods for Efficient Neural Network Inference, Aug. 2021.
[arXiv:2103.13630](https://arxiv.org/abs/2103.13630).
- 📄 A. Graves, S. Fernández, F. Gomez, J. Schmidhuber.
Connectionist temporal classification: labelling unsegmented sequence data with recurrent neural networks.
In *Proc. ICML*, pp. 369–376, Pittsburgh, PA, June 2006.
- 📄 A. Graves.
Sequence Transduction with Recurrent Neural Networks.
In *Proc. ICML*, Edinburgh, Scotland, June 2012.
Workshop on Representation Learning, [arXiv:1211.3711](https://arxiv.org/abs/1211.3711).

- 📄 A. Gulati, J. Qin, C.-C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu, R. Pang.
Conformer: Convolution-Augmented Transformer for Speech Recognition.
In *Proc. Interspeech*, pp. 5036–5040, Shanghai, China, Oct. 2020.
- 📄 T. Hennen, L. Brackmann, T. Ziegler, S. Siegel, S. Menzel, R. Waser, D. J. Wouters, D. Bedau.
Synaptogen: A Cross-Domain Generative Device Model for Large-Scale Neuromorphic Circuit Design.
IEEE Transactions on Electron Devices, Vol. 71, No. 9, pp. 5345–5353, Sept. 2024.
arXiv:2404.06344.
- 📄 V. Panayotov, G. Chen, D. Povey, S. Khudanpur.
Librispeech: an ASR Corpus Based on Public Domain Audio Books.
In *Proc. IEEE ICASSP*, pp. 5206–5210, Queensland, Australia, April 2015.

- 📄 N. Rossenbach, B. Hilmes, L. Brackmann, M. Gunz, R. Schlüter.
Running Conventional Automatic Speech Recognition on Memristor Hardware:
A Simulated Approach.
In Proc. Interspeech, Rotterdam, The Netherlands, Aug. 2025.
[arXiv:2505.24721](#).
- 📄 D. B. Strukov, G. S. Snider, D. R. Stewart, R. S. Williams.
The Missing Memristor Found.
Nature, Vol. 453, pp. 80–83, May 2008.